

Estimating Just-Noticeable Distortion for Video

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Abstract—Just-noticeable distortion (JND), which refers to the maximum distortion that the human visual system (HVS) cannot perceive, plays an important role in perceptual image and video processing. In comparison with JND estimation for images, estimation of the JND profile for video needs to take into account the temporal HVS properties in addition to the spatial properties. In this paper, we develop a spatio-temporal model estimating JND in the discrete cosine transform domain. The proposed model incorporates the spatio-temporal contrast sensitivity function, the influence of eye movements, luminance adaptation, and contrast masking to be more consistent with human perception. It is capable of yielding JNDs for both still images and video with significant motion. The experiments conducted in this study have demonstrated that the JND values estimated for video sequences with moving objects by the model are in line with the HVS perception. The accurate JND estimation of the video towards the actual visibility bounds can be translated into resource savings (e.g., for bandwidth/storage or computation) and performance improvement in video coding and other visual processing tasks (such as perceptual quality evaluation, visual signal restoration/enhancement, watermarking, authentication, and error protection).

Index Terms—Contrast sensitivity function (CSF), discrete cosine transform (DCT), eye movement, just-noticeable distortion (JND).

I. INTRODUCTION

A MAIN aim of designing visual communication systems is to use the least resources to achieve the highest visual quality with respect to certain constraints such as bit rate, complexity, and maximum delay. In most circumstances, the human visual system (HVS) makes final evaluations on the quality of images and video that are processed, transmitted, and displayed. Thus, it is essentially futile to spend significant effort on encoding those signals that are beyond the human perception. Just noticeable distortion (JND), which accounts for the maximum distortion that the HVS does not perceive, can serve as a perceptual threshold to guide an image/video processing task. In image compression schemes, JND can be used to optimize the quantizer [1]–[3] or to facilitate the rate-distortion control [3]. Information of higher perceptual significance is given more bits and preferentially encoded, so that the resultant image is more appealing. In video compression schemes, JND plays more diverse roles. As in image compression, JNDs for video can be used to

improve quantizers [4] and bit allocation; moreover, motion estimation can be facilitated with the help of JND profiles [5]. For both image and video, objective quality evaluation based on the characteristics of the HVS can be achieved by using the JND [6]–[10].

JND estimation for images has been relatively well developed. Pixel-based JND models such as the ones proposed in [2], [11], and [12] basically take into account two components: *luminance adaptation* and *contrast masking*. In [2], the maximum effect between luminance adaptation and contrast masking is used for JND estimation, while in [11] luminance adaptation is regarded as the major factor affecting JND. In [12], both luminance adaptation and contrast masking are combined for a more general pixel-based JND model. In the subband domain, the spatial contrast sensitivity function (CSF), luminance adaptation, and contrast masking can be incorporated into a JND model [1], [2], [13], [14]. Based on spatial CSF, an early DCT-based scheme for estimating the perceptual threshold is presented in [13]. This scheme was subsequently improved into the DCTune model [1] after the luminance adaptation effect was added to the base threshold and contrast masking [15] was used as the elevation factor. More recently, the DCTune model has been modified [3] to work with foveal regions instead of a single pixel. In order to incorporate the contrast masking effect into vision model, block classification for different local structures was introduced in [16]. In [14], more realistic luminance adaptation was considered for digital images to better fit the empirical parabola curve of the human perception [17].

Temporal characteristics of the HVS can be modeled for video. The majority of the related work has been devoted to evaluation of perceptual error between an original video sequence and its processed version [6]–[9], [18], without explicit mathematical expressions for JND. In fact, JND is a property of the video itself even when the visual signal is not processed. Therefore, it is meaningful to derive an explicit equation for evaluating the JND of any frame in video based on the temporal characteristics of the HVS. Furthermore, a *stand-alone* JND estimator for video signals would facilitate wider and/or more convenient applications in visual processing at different constraints. There are two existing JND estimators for video in pixel domain [4], [11]. In [4], the average luminance difference between successive frames is used as the temporal inconsistency to evaluate the JND values, while, in [11], activity variations in successive frames are used for forming the JND profile. Both the aforementioned models approximately use temporally varying pixel values of video for JND estimation.

There is no reported explicit JND model for video in subbands to account for all major factors, i.e., spatial CSF, temporal CSF, luminance adaptation, and contrast masking. A spatio-temporal CSF model has been devised by Kelly [19] from his experiments on visibility thresholds under stabilized viewing con-

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ditions. Daly [20] extended Kelly's model to fit unconstrained natural viewing conditions with a consideration of eye movements. The tracking of moving objects by human eyes alleviates the blurring effect of motion [21], [22]. However, Daly did not work out a comprehensive HVS model for video that considers other factors affecting JND, such as luminance adaptation and contrast masking.

In this paper, we propose a method for explicit JND estimation in the DCT domain, with integrated formulation for both images and video. The model incorporates spatio-temporal CSF, eye movements, luminance adaptation, and intra- and inter-band contrast masking. This paper is organized as follows. The base visibility threshold is derived with the eye-movement-compensated spatio-temporal CSF model in Section II. By fitting the model with the psychophysical data in [13], the model parameters have been determined. In Section III, luminance adaptation and contrast masking based on block classification are incorporated to accomplish the complete JND estimator. We present and discuss the experimental results in Section IV and conclude this paper in Section V.

II. BASE THRESHOLD IN DCT SUBBANDS

Here, we compute the maximum error that can be tolerated for each DCT coefficient with consideration of spatio-temporal CSF (i.e., the relation of visibility threshold with spatial and temporal frequencies of visual signal).

A. Baseline Spatio-Temporal CSF Model

The HVS has evolved to be a highly complex mechanism for visual perception. In psychophysical experiments, visual acuity varies with both spatial and temporal frequencies of the stimuli. The HVS is sensitive to the contrast and can only sense a signal whose contrast is above a certain threshold with respect to the signal frequency. The reciprocal of this threshold is the *contrast sensitivity* which is a function of spatial and temporal frequencies. Spatio-temporal contrast sensitivity is sometimes referred to as the spatial acuity of the HVS depending on the velocity of the image traveling across the retina, where the retinal image velocity implicitly denotes the temporal frequency. The spatio-temporal contrast sensitivity function (CSF) is shown in Fig. 1. The corresponding baseline sensitivity based upon the work of Kelly [19] and Daly [20] is as follows:

$$G(\alpha, \nu) = c_0 \left(k_1 + k_2 |\log(\varepsilon \cdot \nu / 3)|^3 \right) \cdot \nu \cdot \alpha^2 \exp(-c_1 \cdot \alpha \cdot (\varepsilon \cdot \nu + 2) / k_3) \quad (1)$$

where ν (degrees per second) is the retinal image velocity and $\alpha/2\pi$ (cycles per degree) denotes the spatial frequency; the empirical constants k_1 , k_2 , and k_3 are set as 6.1, 7.3, and 23 [19], respectively, $\varepsilon = 1.7$, and c_0 and c_1 control the magnitude and the bandwidth of a CSF curve (see Fig. 1) and can be decided with subjective viewing test data for the subband decomposition in use. It should be noted that luminance dependency is not considered in (1).

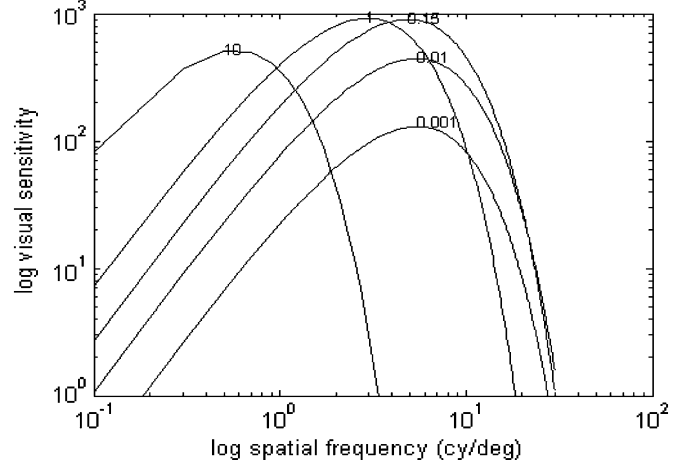


Fig. 1. Spatial CSF at different retinal velocities (0.01, 0.001, 0.15, 1, and 10 deg/s) [20].

If we consider the (i, j) th subband in the n th DCT block in the t th frame, the corresponding CSF can be derived from (1) as

$$G(n, i, j, t) = c_0 \left(k_1 + k_2 |\log(\varepsilon \cdot \nu(n, t) / 3)|^3 \right) \cdot \nu(n, t) \cdot (2\pi\rho_{i,j})^2 \cdot \exp(-2\pi\rho_{i,j} \cdot c_1 \cdot (\varepsilon \cdot \nu(n, t) + 2) / k_3) \quad (2)$$

where $i, j = 0, 1, \dots, N-1$, and N is the dimension of the DCT block, $\nu(n, t)$ depicts the associated retinal image velocity, and $\rho_{i,j}$ (cycles per degree) is the spatial subband frequency

$$\rho_{i,j} = \frac{1}{2N} \sqrt{(i/\omega_x)^2 + (j/\omega_y)^2} \quad (3)$$

where ω_x and ω_y are the horizontal and vertical sizes of a pixel in degrees of visual angle, respectively. They are related to the viewing distance ℓ and the display width Λ of a pixel on the monitor, as follows:

$$\omega_h = 2 \cdot \arctan\left(\frac{\Lambda_h}{2 \cdot \ell}\right), \quad h = x, y. \quad (4)$$

B. Eye-Movement Effect

For a practical formulation of the spatio-temporal CSF, the influence of observers' eye movement needs to be taken into account for motion imagery [9], [21], [22], and the retinal velocity in (2), $\nu(n, t)$, has to be expressed by easily measurable variables from the moving images. Fast motion blurs the HVS perception; the spontaneous *smooth-pursuit eye movement* (SPEM) tends to track the moving object and reduce the retinal velocity of the image, and thus compensates the loss of sensitivity due to motion. Besides the SPEM, two additional types of eye movements are reported: the *natural drift eye movement* and the *saccadic eye movement* [23]. The former is very slow (0.8–1.5 deg/s) and responsible for the perception of static images [24], while the latter is responsible for rapidly moving objects to which the HVS sensitivity is very low. Because of the

eye movements, the perceived velocity at the retina (retinal velocity) is different from the image plane velocity that can be usually obtained through motion estimation.

Based on the heuristic eye movement model devised by Daly [20], the retinal image velocity can be expressed as

$$\nu(n, t) = \nu_I(n, t) - \nu_E(n, t) \quad (5)$$

where $\nu_I(n, t)$ represents the image plane object velocity in retina if no eye motion were involved; $\nu_E(n, t)$ is the eye movement velocity to be determined as

$$\nu_E(n, t) = \min[(g \cdot \nu_I(n, t)) + \nu_{\min}, \nu_{\max}] \quad (6)$$

where g is the gain of the SPEM which indicates the efficiency of object tracking, $\nu_{\min}$ is the minimum eye velocity due to the drift movement, $\nu_{\max}$ is the maximum eye velocity before saccadic movement. Equations (5) and (6) convert the image plane velocity $\nu_I(n, t)$ to the retinal image velocity $\nu(n, t)$. The parameter g is assigned with a value of 0.92, and $\nu_{\min}$ and $\nu_{\max}$ are set to 0.15 and 80.0 deg/s.

We obtain the image velocity with a motion estimation technique (e.g., block matching [25] or optical flow calculation [26]). Let $(MV_x(n, t), MV_y(n, t))$ be the motion vector for the n^{th} block in the t^{th} frame; then the image velocity is

$$\nu_I(n, t) = f \cdot \sqrt{(MV_x(n, t) \cdot \omega_x)^2 + (MV_y(n, t) \cdot \omega_y)^2} \quad (7)$$

where f is the frame rate (in *frames per second*) of video, and $\nu_I(n, t)$ is measured in *degrees of visual angle per second*.

When $\nu(n, t) \equiv 0.15$ deg/s (i.e., only natural drift movement occurs), the spatio-temporal CSF is equivalent to the spatial (static) CSF [20], so the formulas derived in this paper is also applicable for images if the said value of $\nu(n, t)$ is used.

C. Base Distortion Threshold

When (2) is used for predicting distortion threshold due to spatio-temporal CSF, several factors need to be considered: 1) the sensitivity modeled by (2) represents the inverse of distortion threshold; 2) the CSF-derived threshold expressed in luminance needs to be scaled to grey levels for digital images [13]; 3) since (2) is derived from the experimental data of one-dimensional spatial frequencies (i.e., i or j is equal to 0), for an arbitrary subband ($i \neq 0$ and $j \neq 0$), the threshold is actually higher than the one given by (2), and therefore a compensating term (as the last term of the right-hand side in (8) below) needs to be introduced [13] for a DCT subband. With all considerations mentioned above, the base threshold for a DCT subband is determined as

$$T(n, i, j, t) = \frac{1}{G(n, i, j, t)} \cdot \frac{M}{\phi_i \phi_j (L_{\max} - L_{\min})} \cdot \frac{1}{r + (1 - r) \cdot \cos^2 \theta_{i,j}} \quad (8)$$

where $L_{\max}$ and $L_{\min}$ represent the display luminance values corresponding to the maximum and minimum grey levels, M

is the number of gray levels (256 in most imagery systems); ϕ_i and ϕ_j are DCT normalization factors

$$\phi_u = \begin{cases} \sqrt{\frac{1}{N}}, & u = 0 \\ \sqrt{\frac{2}{N}}, & u \neq 0 \end{cases} \quad u = 0, 1, \dots, N-1 \quad (9)$$

where r is set to 0.6 [13], and $\theta_{i,j}$ accounts for the effect of an arbitrary subband [the factor 3) mentioned above]

$$\theta_{i,j} = \arcsin \frac{2\rho_{i,0}\rho_{0,j}}{\rho_{i,j}^2}. \quad (10)$$

When $i = 0$ or $j = 0$, $\theta_{i,j} = 0$ and $T(n, i, j, t)$ takes its smallest value since $r + (1 - r) \cos^2 \theta_{i,j} = 1$; when $i = j$, $\theta_{i,j} = 90^\circ$ and $T(n, i, j, t)$ takes its biggest value since $r + (1 - r) \cos^2 \theta_{i,j} = r$.

D. Determination of c_0 and c_1

In (2), c_0 and c_1 regulate the value level and the bandwidth of the CSF respectively. With $\nu(n, t) = 0.15$ deg/sec in (8), we estimated c_0 and c_1 by fitting the resultant $T(i, j)$ to the spatial CSF model in [13]. Only the ac components of the upper-left triangle in the DCT block are used for fitting because of the symmetry of the spatial frequencies expressed as (3). The visibility thresholds t_s are calculated for each DCT frequency ρ_s using the JND model in [13]. We use the least mean squares (LMS) approach to get the best-fitted c_0 and c_1 for the data pair (ρ_s, t_s) as

$$(c_0, c_1) = \arg \min \left\{ \sum_s [t_s - T(\rho_s, c_0, c_1)]^2 \right\} \quad (11)$$

where $T(\rho_s, c_0, c_1)$ is the threshold [derived from (8)]. As a result, $c_0 = 7.126$ and $c_1 = 0.565$.

III. LUMINANCE ADAPTATION AND CONTRAST MASKING

For a complete spatio-temporal CSF model, we need to include luminance adaptation and, as a comprehensive JND estimator, different contrast masking has to be considered. For luminance adaptation, the response of the HVS depends on the luminance of the surroundings. Intra-band masking refers to the reduction in the distortion visibility in a subband due to the signal in that subband itself, while inter-band masking refers to the reduction in the distortion visibility in a subband at the presence of visual components in other subbands. In this study, inter-band masking is evaluated on a block basis.

The complete JND estimator is modeled as

$$\text{JND}(n, i, j, t) = T(n, i, j, t) \cdot a_{\text{Lum}}(n, t) \cdot a_{\text{intra}}(n, i, j, t) \cdot a_{\text{inter}}(n, t) \quad (12)$$

where $a_{\text{Lum}}(n, t)$, $a_{\text{intra}}(n, i, j, t)$, and $a_{\text{inter}}(n, t)$ account for the effects of luminance adaptation, intra-band masking, and inter-band masking, respectively.

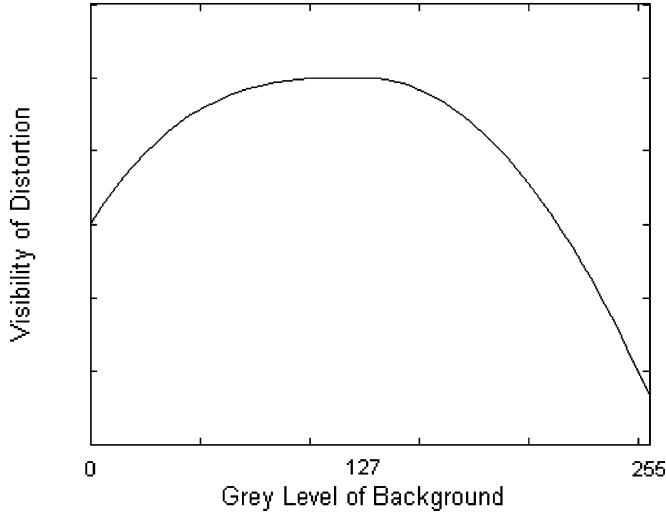


Fig. 2. Distortion visibility as a function of background brightness [2].

A. Luminance Adaptation

For digital images, the HVS visibility threshold is higher in dark and light regions, i.e., the HVS is more sensitive to the noise in medium gray–gray regions, as shown in Fig. 2.

Since the average local intensity can be represented by its dc component $C(n, 0, 0, t)$, the luminance adaptation factor for the n DCT block is determined as [14]

$$a_{\text{lum}}(n, t) = \begin{cases} k_1 \left(1 - \frac{2 \cdot C(n, 0, 0, t)}{M \cdot N}\right)^{\lambda_1} + 1, & \text{if } C(n, 0, 0, t) \leq \frac{M \cdot N}{2} \\ k_2 \left(\frac{2 \cdot C(n, 0, 0, t)}{M \cdot N} - 1\right)^{\lambda_2} + 1, & \text{otherwise} \end{cases} \quad (13)$$

where $k_1 = 2$, $k_2 = 0.8$, $\lambda_1 = 3$, and $\lambda_2 = 2$ [14].

B. Intra- and Inter-Band Contrast Masking

The extent of contrast masking depends on the local intensity activity of the image. The HVS sensitivity to error is generally higher in smooth (or plain) regions and lower in the texture region [14], [16], while the sensitivity for edge region lies in between. Therefore, contrast masking should be discriminated for different image contexts.

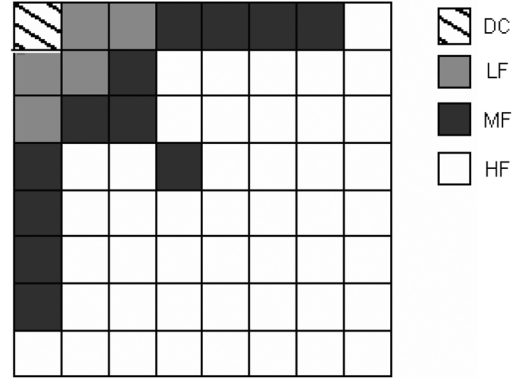


Fig. 3. Block classification scheme for a DCT block [16].

We follow the method in [14] and [16] to handle inter-band masking. A DCT block can be divided into four parts: dc, low-frequency (LF), medium-frequency (MF), and high-frequency (HF) parts, as shown in Fig. 3. If we denote the sums of the absolute DCT coefficients in LF, MF, and HF with L , M , and H , then the block texture energy $TexE$ is calculated as

$$TexE = M + H. \quad (14)$$

According to L , M , H , and $TexE$, blocks are classified as *Texture*, *Edge*, and *Plain* ones [14], [16]. Then inter-band masking can be represented as (15), shown at the bottom of the page, where $\xi_1 = 290$, $\xi_2 = 900$, $\delta_1 = 1.25$, and $\delta_2 = 1.125$.

As for intra-band masking, we modify the model in [14] to exclude intra-band masking from taking effect in LF and MF subbands for nontextured blocks, as shown in (16) at the bottom of the page, where $C(n, i, j, t)$ is the DCT coefficient, and $\varepsilon = 0.36$ [2], [14].

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

To evaluate the performance of JND models, the generated JND profiles can be used to guide noise shaping in an image or video [2], [5], [14], and appropriate subjective viewing tests are then conducted to assess the quality of the resultant visual signal. The proposed JND estimator has been compared with Zhang *et al.*'s JND model [14] (referred to as Model I hereinafter) and Daly's model¹ [20] (referred to as Model II hereinafter) in the experiments. These two models are the most rel-

¹With c_0 and c_1 determined in Section II-D, because Daly's original model is not for DCT domain.

$$\alpha_{\text{inter}}(n, t) = \begin{cases} 1 + [(TexE(n, t) - \xi_1) / (2\xi_2 - \xi_1)] \delta_1, & \text{for TEXTURE block} \\ \delta_1, & \text{for EDGE block and } L + M > 400 \\ \delta_2, & \text{for EDGE block and } L + M \leq 400 \\ 1, & \text{for PLAIN block} \end{cases} \quad (15)$$

$$a_{\text{intra}}(n, i, j, t) = \begin{cases} 1, & \text{for } (i, j) \in LF \cup MF \text{ in smooth \& edge block} \\ \max \left\{ 1, \left[\frac{C(n, i, j, t)}{T(n, i, j, t) \cdot a_{\text{lum}}(n, t)} \right]^\varepsilon \right\}, & \text{otherwise} \end{cases} \quad (16)$$



(a)



(b)



(c)

Fig. 4. Noise-injected *Lena* images with Model I, Model II, and the proposed JND model. (a) Model I, PSNR: 31.25 dB. (b) Model II, PSNR: 33.08 dB. (c) the proposed JND model, PSNR: 31.09 dB.

evant existing models for comparison, since Model I is the recent enhancement of the well-acknowledged models [1], [13] in DCT subbands, while Model II is an enhanced version of [19] for spatio-temporal CSF. The relevant parameters of the models are determined under the same experimental conditions: a CRT monitor (17-in Philips 107X₂) where ω_x and ω_y are equal to 0.0342° , and ℓ is 50 cm. When a model (Model II or the proposed model) is used for images in the following experiments, $\nu(n, t)$ is set to 0.15 deg/s.



Fig. 5. Images for the experiments. (a) Cameraman 256×256 . (b) Pepper 512×512 . (c) Goldhill 256×256 . (d) Airplane 512×512 . (e) Lena 512×512 . (f) Baboon 256×256 . (g) Barbara 512×512 . (h) Girl 256×256 . (i) House 512×512 . (j) Splash 512×512 .

The noise injection can be described as

$$C_{\text{noise}}(n, i, j, t) = C(n, i, j, t) + M_{n,i,j}^{\text{random}} \cdot J(n, i, j, t) \quad (17)$$

where $C_{\text{noise}}(n, i, j, t)$ is the noise-injected DCT coefficient, $M_{n,i,j}^{\text{random}}$ takes +1 or -1 randomly, and $J(n, i, j, t)$ represents the JND obtained via a model (Model I, Model II, or the proposed model). A JND model can avoid yielding values larger than the actual HVS thresholds via adjusting its overall gain. If a model derives better JNDs, (17) allows higher injected-noise energy (measured by PSNR) without jeopardizing picture quality.

The subjective viewing tests were conducted in a room illuminated by fluorescent ceiling lights, and these are typical conditions under which people would view digital images. The subjects' ages were 21–24, and their eyesight was either normal or had been corrected to be normal with spectacles. In the following subsections, more details will be given for the respective subjective viewing experiments.

A. Evaluation on Images

For images, the proposed model is similar to Model I and differs from Model II in that the latter does not exploit the content-based properties of an image (i.e., luminance adaptation and contrast masking).

Fig. 4 shows the noise-contaminated *Lena* images by Model I, Model II, and the proposed model. We find that the noise injected into the images is hardly noticeable, so the three models yield values not larger than the actual HVS thresholds. As can be seen, the proposed model yields the lowest PSNR (about 2 dB lower than that for Model II), and this reflects that the proposed model is able to exploit the HVS bounds more aggressively without introducing noticeable visual distortions.

For a comprehensive evaluation, we have conducted subjective viewing tests to give quantitative scores for all images (with different visual contents and spatial complexity) shown in Fig. 5 with JND-guided noise shaping. In each viewing test, two images of the same scene (i.e., the original image and its noise-contaminated version) were juxtaposed on the screen, and six subjects (three were working in the image processing field and three were naive) were employed. In each session of the experiment, subjects viewed two images and then took time to vote on the comparative quality of the images, using the continuous quality comparison scale shown in Table I. The subjects were not allowed to respond until they had viewed the images for at least

TABLE I
SUBJECTIVE RATING CRITERION FOR COMPARATIVE
VISUAL QUALITY OF AN IMAGE PAIR

Subjective score	Description
-3	the right one is much worse than the left one
-2	the right one is worse than the left one
-1	the right one is slightly worse than the left one
0	the right one has the same quality as the left one
1	the right one is slightly better than the left one
2	the right one is better than the left one
3	the right one is much better than the left one

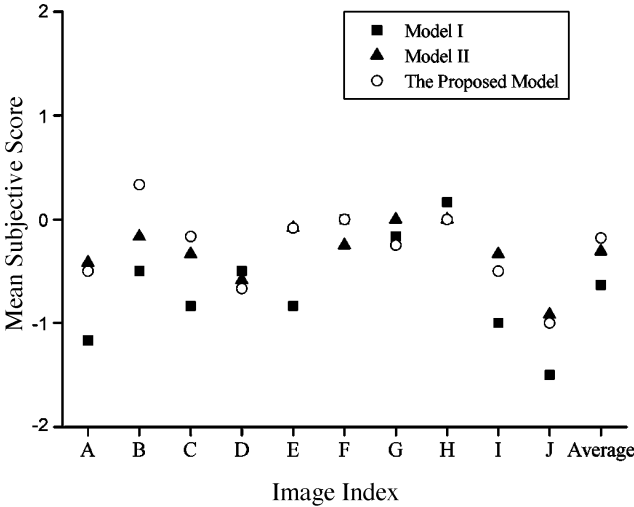


Fig. 6. Mean subjective scores for the noise-injected images with the three JND models.

TABLE II
STANDARD DEVIATIONS OF THE SUBJECTIVE SCORES FOR THE
NOISE-INJECTED IMAGES

Image	A	B	C	D	E	F	G	H	I	J
Model I	0.26	0.55	0.41	0.55	0.41	0	0.41	0.41	0.55	0.45
Model II	0.49	0.41	0.52	0.49	0.2	0.42	0	0	0.41	0.2
The Proposed Model	0.55	0.52	0.41	0.52	0.49	0	0.42	0	0.55	0

two seconds. The order of the presentation of the image pairs was randomized in each session, and the noised image appeared randomly on the left- or right-hand side of the screen. The results indicated no notable difference on whether a subject has image processing knowledge or not.

Fig. 6 shows the mean subjective score by all subjects for each noise-injected image with one of the JND models, while Table II lists the corresponding standard deviations. A negative subjective score indicates that the noise-injected image has worse perceptual quality than the original image, and its magnitude represents the extent of quality degradation. The slightly higher average quality score (in difference of 0.33 in Fig. 6) with Model II than Model I is due to the somewhat excessive intra-band masking in LF and MF subbands for nontextured blocks in Model I.² As can be observed, the proposed model leads to similar visual quality on average as Model II in the noise-injected images and slightly better quality on average than

²We have remedied this by excluding intra-band masking from taking effect in LF and MF subbands for nontextured blocks [via (16)].

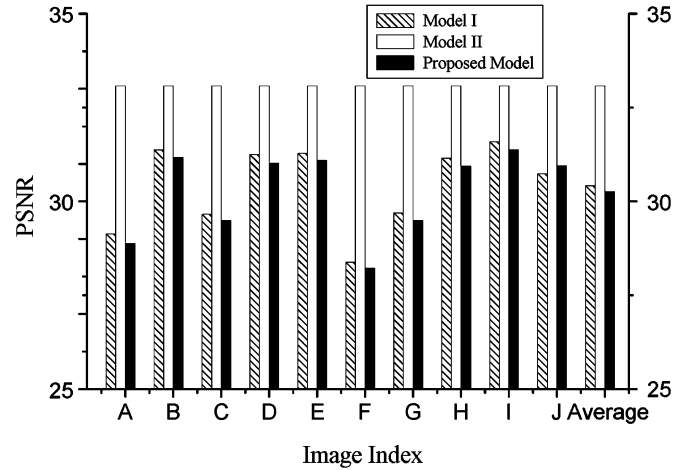


Fig. 7. PSNRs of noise-injected images by the three models.

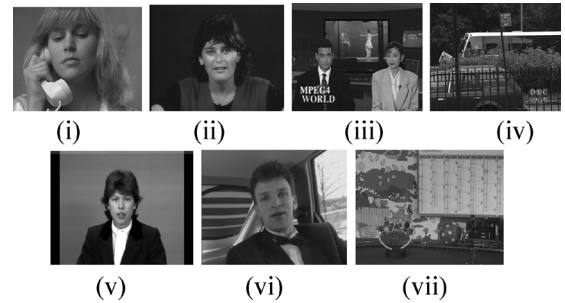


Fig. 8. Videos for the experiments. (i) *Susie.cif*. (ii) *Miss American.cif*. (iii) *News.qcif*. (iv) *Bus.cif*. (v) *Claire.cif*. (vi) *Carphone.qcif*. (vii) *Caltrain.cif*.

Model I. Overall, the subjective scores resulting from the three models in Fig. 6 are very similar, because the score variations among the three are insignificant (the average range of subjective score variations is below 0.45).³

The PSNR has been used to measure the visual content variations after the JND-guided noise injection. At the same level of perceived picture quality, a better JND model yields more aggressive JNDs (i.e., resulting in lower PSNR). Fig. 7 shows the PSNRs of the noise-injected images using the three models. The PSNRs (as well as the average PSNR) for the images with the proposed model are similar to those for Model I. As aforementioned, Model II for images is image-independent, and therefore the associated PSNR remains the same for every image. The proposed model yields more aggressive JNDs than Model II towards the actual HVS thresholds, with the evidence of an average PSNR reduction of 2.82 dB from Model II (as shown in Fig. 7 for all ten images), without jeopardy of visual quality (see Fig. 6).

B. Evaluation on Video

We have conducted experiments to test how the models work for video sequences. Seven sequences (*Susie*, *Miss American*, *News*, *Bus*, *Claire*, *Carphone*, and *Caltrain*), as shown in Fig. 8, with different visual contents and object motion) are distorted by inserting the noise according to (17), for the experiments. The proposed model should outperform the other two models in

³In comparison with the scale of the subjective scores.

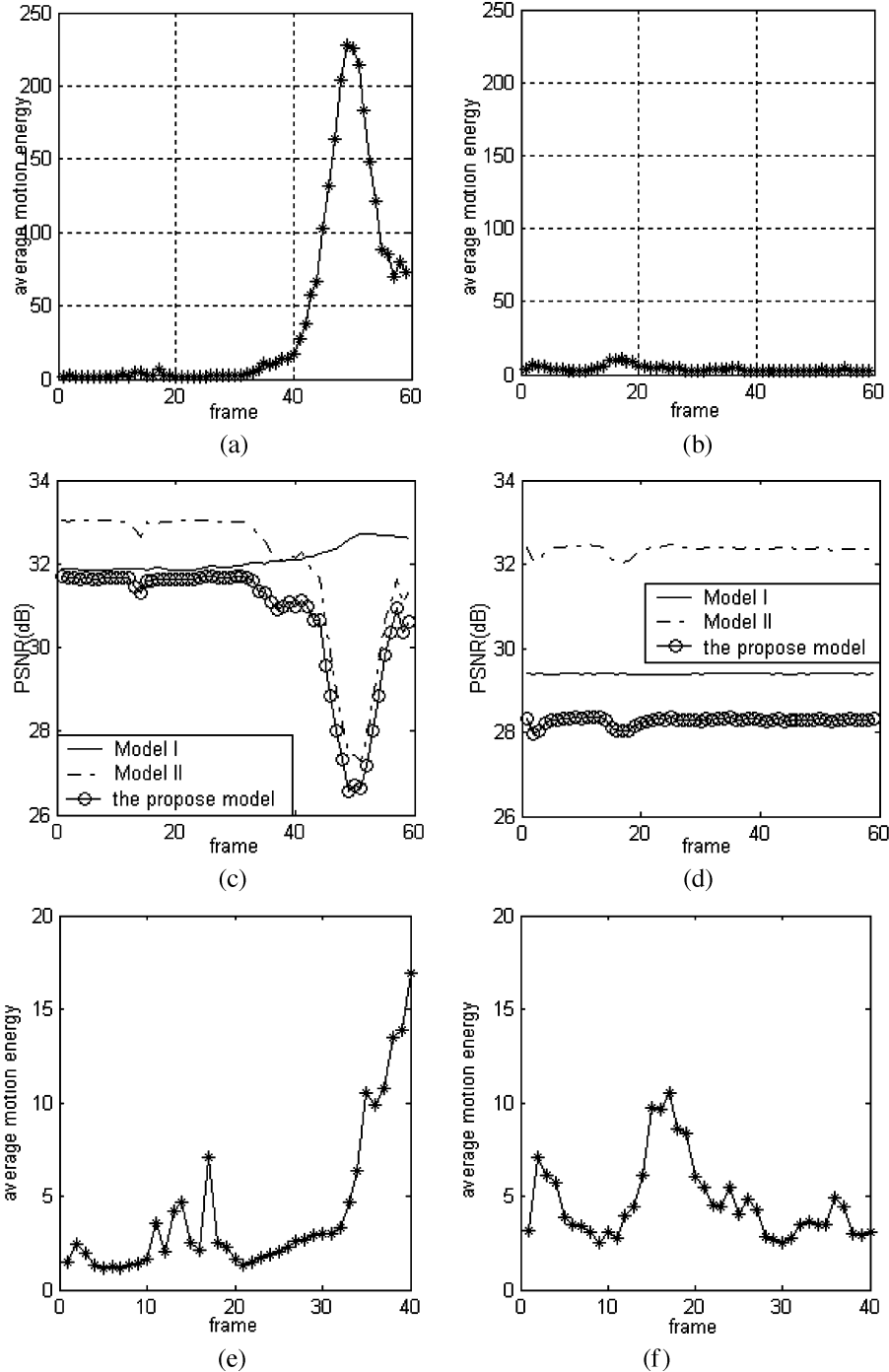


Fig. 9. Demonstration of motion effect. (a) *Susie*: average motion energy versus frame. (b) *Miss American*: average motion energy versus frame. (c) *Susie*: PSNR versus frame. (d) *Miss American*: PSNR versus frame. (e) Details of (a) before frame 40. (f) Details of (b) before frame 40.

video, since Model I does not exploit temporal CSF while Model II does not exploit the content-based properties of an image.

1) *Effect of Motion*: Fig. 9 illustrates how the models perform in the presence of motion for the noise-injected *Susie* and *Miss American* sequences. First, we measure the motion of each frame using the *average motion energy*, which is defined as

$$Avg_MotionEnergy(t) = \frac{1}{N_f} \sum_n (MV_x^2(n, t) + MV_y^2(n, t)) \quad (18)$$

where N_f is the number of blocks in a frame.

Fig. 9(a) and (b) denotes the *average motion energy* over the frames for the noise-injected *Susie* and *Miss American* video sequences, with the three JND models, while (e) and (f) illustrate the zoom-in details of (a) and (b) before Frame 40. Fig. 9(c) and (d) shows the PSNRs over the frames for the two noise-injected sequences.

As shown in Fig. 9(a) and (e), small motion and big motion take place around frames 20 and 40, respectively, in the noise-injected *Susie* sequence; as expected, Fig. 9(c) shows the corresponding small drop and big drop of PSNR for Model II and the proposed model. Similar phenomena occur in *Miss American*

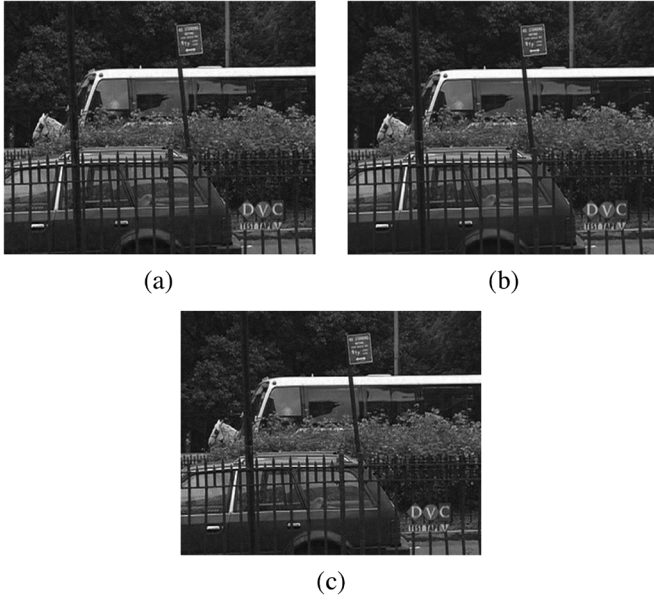


Fig. 10. Noise-injection to the first frame of *Bus* sequence with Model I, Model II, and the proposed JND model. (a) Model I, PSNR: 28.93 dB. (b) Model II, PSNR: 33.08 dB. (c) The proposed JND model, PSNR: 28.70 dB.

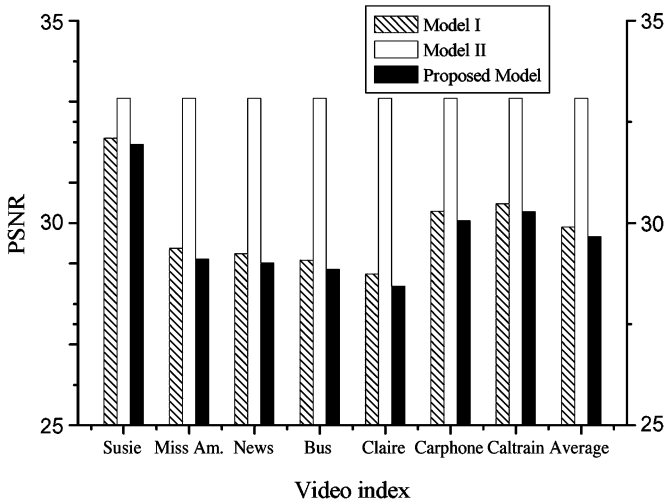


Fig. 11. PSNRs of noise-contaminated frames of videos by the three models (without temporal CSF effect).

when there is small motion around frames 3 and 18, as illustrated in Fig. 9(b), (f), and (d).

It is obvious that Model I is not able to respond to motion since the temporal CSF has not been incorporated. The proposed JND model predicts correctly (inheriting the temporal characteristics from Model II) that more distortion can be tolerated with higher motion.

We notice that the motion energy actually fluctuates in Fig. 9(e) and (f),⁴ while the PSNR curves by Model II and the proposed model in Fig. 9(c) and (d) do not reflect such motion fluctuations. In fact, this demonstrates the effect of eye movement: for low motion, the eye movement can compensate some, if not all loss of sensitivity, and thus the JNDs derived by

⁴This is apart from the aforementioned motion around frames 20 and 40 in Fig. 9(e) and frames 3 and 18 in Fig. 9(f).

Model II and the proposed model do not undertake considerable elevation for these slight motion fluctuations.

In addition, the PSNR curve by the proposed model has similar shape but with a lower valuation level than that by Model II; this demonstrates the effectiveness of luminance adaptation and contrast masking considered in the proposed model.

2) *Noise Shaping for Individual Frames*: In order to provide the ground to demonstrate the temporal CSF effect with the proposed model (to be discussed in Section IV-B3), noise-injection has been performed with the first frame of the video sequences in Fig. 8. Since each frame is treated as a still image, $\nu(n, t)$ is set to be 0.15 deg/s in Model II and the proposed model. In line with the results obtained in Section IV-A, the same subjective viewing tests have confirmed that the three models yield similar perceptual quality in the noise-injected frames. Fig. 10 shows the noise-injected images for the first frame of *Bus* sequence with Model I, Model II, and the proposed JND model.

Fig. 11 illustrates the PSNRs of noise-contaminated frames of different videos by the three models (without temporal CSF effect). Similar to the results in Section IV-A, the proposed model performs slightly better than Model I in PSNR reduction, and it yields an average of 3.41 dB of PSNR reduction from Model II (as shown in Fig. 11).

3) *Overall Performance*: For a comprehensive performance comparison, we followed the procedures of the Double Stimulus Continuous Quality Scale (DSCQS) method in Rec. ITU-R BT.500 [27], in the subjective experiments for video. Fig. 12 shows the presentation course. The *Mean Opinion Score* (MOS) scales are adopted for quality grading: Excellent (100–80), Good (80–60), Fair (60–40), Poor (40–20), and Bad (20–0), for both an original sequence and the associated processed sequence. Ten viewers (three were in the image processing field and seven were naive) were required to give MOS for both the original and the processed sequences (they were not told which sequence was the original one). Then, the different MOS (DMOS) is obtained by subtracting the MOS of the processed sequence from that of the original one. A higher DMOS indicates bigger quality degradation of the processed sequence. Again, there is no notable difference found regarding whether a subject has image processing knowledge.

Fig. 13 compares the mean DMOS with the ten viewers for each sequence, and Table III lists the corresponding standard deviations. It is shown in Fig. 13 that the subjective quality of the noise-injected video sequences is quite similar with the three models (the average DMOSs for Model I, Model II, and the proposed model are 10.7, 10.1, and 10.5, respectively).⁵ However, Fig. 14 demonstrates that the average PSNR of noise-injected sequences by the proposed model is 0.83 dB lower than that for Model I and 3.32 dB lower than that for Model II. Comparing the PSNRs for the proposed model in Figs. 11 and 14, it can be seen that the consideration of temporal CSF brings about 0.6-dB additional perceptual data redundancy on average. In summary, the proposed model is able to give more aggressive JND estimation than the other two models, without affecting the perceptual image quality, and the modeling of temporal effect is effective.

⁵Due to the same reason explained in Section IV-A, the average quality score with Model II is slightly higher than that of Model I (with the difference of 0.6 in Fig. 13).

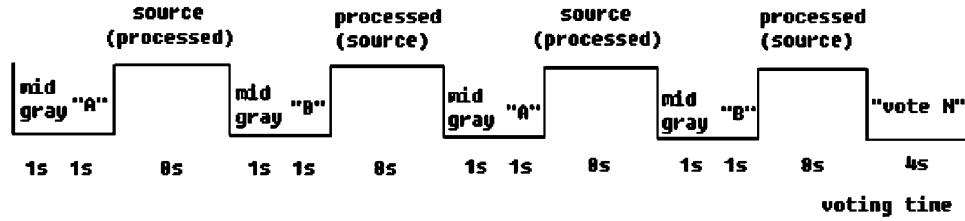


Fig. 12. DSCQS test scheme.

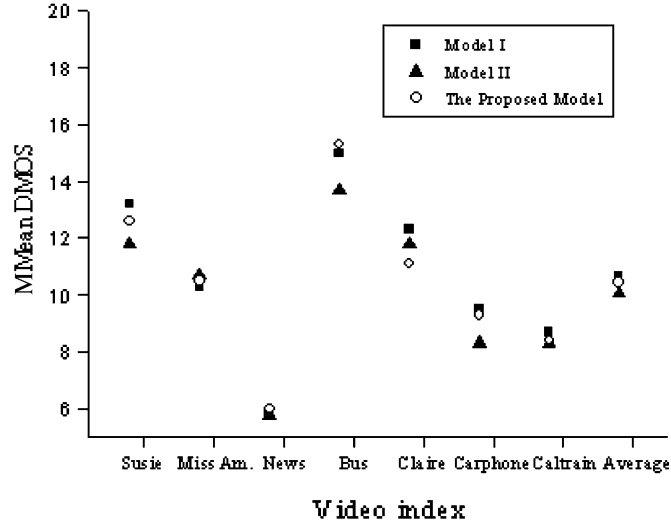


Fig. 13. Mean DMOSs for the noise-injected videos with the three JND models.

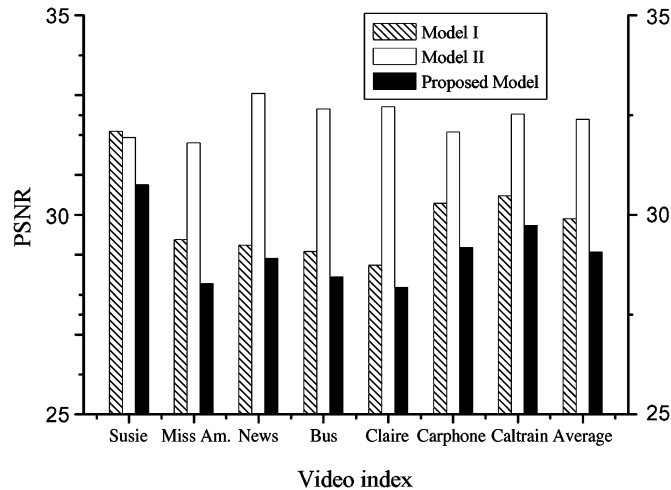


Fig. 14. PSNRs of noise-contaminated videos by the three models.

C. Discussion

A better JND model is capable of determining a more aggressive error profile (e.g., with lower PSNR in the experiments described above) for an image but not causing noticeable visual distortion. Determination of the biggest unnoticeable error bounds of visual signal facilitates various processing tasks for resource savings and performance improvement.

In image and video compression, the budgeted bits can be allocated for better coding quality using a more accurate JND profile, with more bits assigned (via quantization step selection) [2]–[5], [14] to the signal with lower JNDs; only the DCT coefficients above the JNDs need to be coded [28], and this is able

TABLE III
STANDARD DEVIATIONS OF DMOSs FOR THE NOISE-INJECTED VIDEOS

Video	Susie	Miss Am.	News	Bus	Claire	Carphone	Caltrain
Model I	1.93	2.36	1.62	1.63	2.21	2.12	1.42
Model II	1.55	2.11	1.75	2.91	1.93	2.36	2.06
The proposed Model	2.12	2.84	2.11	1.95	1	1.7	2.01

to result in computational complexity reduction and bit savings for more significant signal components (or, alternatively, better perceptual picture quality with a same bit rate).

Accurate JND estimation benefits the noncompression processing processes as well. In watermarking, authentication, and error-protection applications, the accessorial data can be embedded inside the visual signal itself with the guide of the JND profile towards the minimum visual quality degradation [29], [30]. For visual quality/distortion prediction, a metric can be defined or finetuned according to the JND [1], [10], [14] to better match the HVS perception; a JND-based perceptual metric may be also adopted beyond the quality evaluation purpose (e.g., for image synthesis [31]). The optimum edge sharpness may be determined for visual signal enhancement based upon the JNDs [32].

V. CONCLUSION

In comparison with the image case, estimation of JND for video needs to take the temporal HVS properties into account, in addition to the spatial properties. In this paper, a comprehensive, explicit JND estimator has been proposed for video in the DCT domain, to account for the major factors affecting JND, i.e., spatio-temporal CSF, eye movement, luminance adaptation, and intra- and inter-band contrast masking. The integrated equation can be also used for still images with the retinal image velocity being set as a constant.

Experimental results with subjective viewing confirm the improved performance of the proposed model and the effective modeling of temporal effect. The model is capable of predicting more aggressive JND values without introducing noticeable distortion to the original video and, therefore, outperforms the relevant existing models. The more accurate JND estimation towards the actual visibility bounds can facilitate applications in image/video compression, watermarking, perceptual quality evaluation, and visual signal enhancement.

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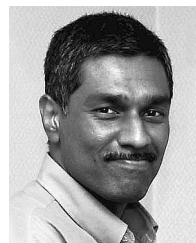
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